IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATIONS 2023

WAVELETS AND APPLICATIONS SOLUTIONS

Answers

SOLUTIONS

1. (a)

$$\hat{X}(z) = \frac{G_0(z)}{2}\left[H_0(z)X(z) + H_0(-z)X(-z)\right] + \frac{G_1(z)}{2}\left[H_1(z)X(z) + H_1(-z)X(-z)\right].$$

Therefore PR condition is:

$$G_0(z)H_0(z) + G_1(z)H_1(z) = 2$$

and

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = 0.$$

(b) i. We have that

$$\frac{1}{\pi}(z + \pi + z^{-1}). + \frac{1}{\pi}(-z + \pi - z^{-1}) = 2.$$

Therefore the half-band condition is satisfied.

ii. We treat $P(z) = \frac{1}{\pi}(z + \pi + z^{-1})$ as a polynomial of degree two and find its roots. The two roots are given by:

$$z_{1,2} = \frac{-\pi \pm \sqrt{\pi^2 - 4}}{2}.$$

Therefore $z_1 = -0.36$ and $z_2 = -2.78$. We need to pick the root inside the unit circle for $G_0(z)$. Therefore

$$G_0(z) = \frac{1}{\sqrt{\pi}}(1 + 0.36z^{-1}).$$

iii. Because of biorthogonality we have:

$$H_0(z) = \frac{1}{\sqrt{\pi}}(z + 2.78).$$

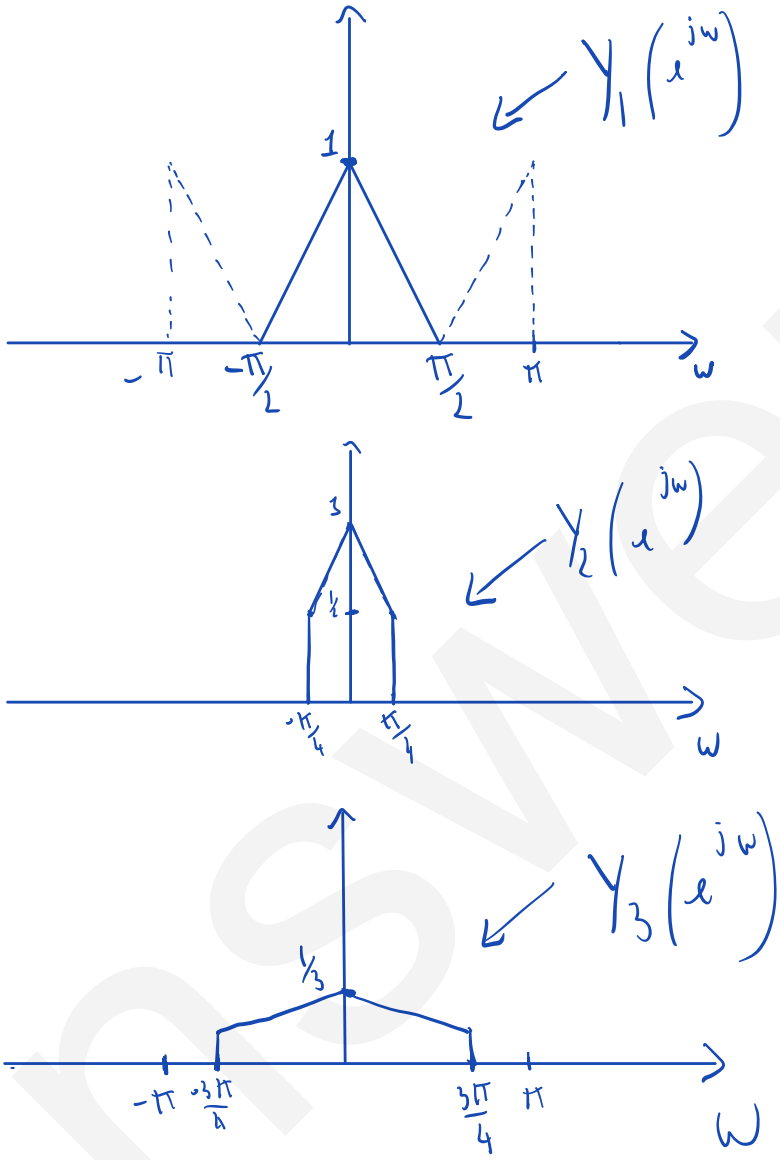
$$G_1(z) = z^{-1}H_0(-z) = \frac{1}{\sqrt{\pi}}(2.78z^{-1} - 1).$$

$$H_1(z) = zG_0(-z) = \frac{1}{\sqrt{\pi}}(z - 0.36).$$

iv. The filter does not satisfy the necessary condition for convergence since $G_0(e^{j\omega}) \neq \sqrt{2}$ for $\omega = 0$ and $G_0(e^{j\omega}) \neq 0$ for $\omega = \pi$. Therefore the limit does not converge and we cannot obtain a continuous-time wavelet from iterating this filter bank.

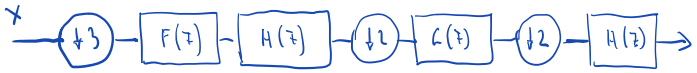
Answers

2. (a) The three spectra are shown below

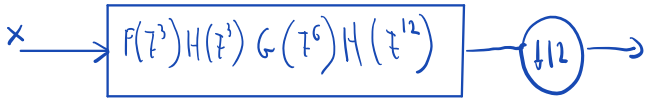


Answers

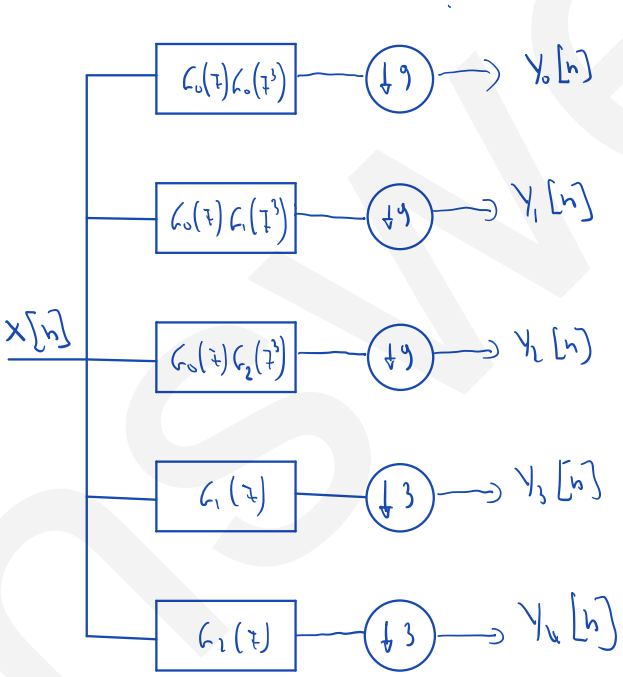
(b) i. The block diagram of the system is:



ii. The equivalent system is:



(c) i. The equivalent filter bank is the following:



ii. $G_1(z)$ has two vanishing moments and $G_2(z)$ has one vanishing moment, so only $y_0[n]$ will be non-zero.

iii. In this case we need two vanishing moments to annihilate the signal so $y_0[n]$,$y_2[n]$ and $y_4[n]$ will be non-zero.

Answers

3. (a) i. The most correct way to show that $\varphi(t+n)$ $n \in \mathbb{Z}$ is an orthonormal system, is to demonstrate that

$$\sum_k |\hat{\varphi}(\omega + 2\pi k)|^2 = 1. \tag{1}$$

Given that the Fourier transform of

$$\varphi(t) = \frac{\sin(\pi t)}{\pi t}$$

is

$$\hat{\varphi}(\omega) = \text{rect}\left(\frac{\omega}{2\pi}\right),$$

then it is clear that condition (1) is satisfied.

ii. By using Poisson summation formula, we have that

$$\sum_n \varphi(t-n) = \sum_k \hat{\varphi}(2\pi k) e^{-j2\pi kt}.$$

Consequently partition of unity is satisfied when

$$\sum_k \hat{\varphi}(2\pi k) e^{-j2\pi kt} = 1.$$

In our case

$$\hat{\varphi}(\omega) = \text{rect}\left(\frac{\omega}{2\pi}\right),$$

therefore, $\hat{\varphi}(0) = 1$ and $\hat{\varphi}(2\pi k) = 0$ when $k \neq 0$. This means that

$$\sum_k \hat{\varphi}(2\pi k) e^{-j2\pi kt} = 1$$

is satisfied.

iii. A. We have the following

$$\begin{aligned}
\sqrt{2}\langle \varphi(t), \varphi(2t-k)\rangle &= 2\textstyle\sum_n g_0[n]\langle \varphi(2t-n), \varphi(2t-k)\rangle \\
&= \textstyle\sum_n g_0[n]\langle \varphi(x-n), \varphi(x-k)\rangle \\
&= g_0[k],
\end{aligned}$$

where the last equality is due to the fact that $\langle \varphi(t), \varphi(t-k)\rangle = \delta_k$.

B. Using Parseval equality we have

$$\begin{aligned}
g_0[n] &= \sqrt{2}\langle \varphi(t), \varphi(2t-k)\rangle \\
&= \tfrac{\sqrt{2}}{4\pi}\langle \hat{\varphi}(\omega), \hat{\varphi}(\omega/2) e^{-j\omega k/2}\rangle \\
&= \tfrac{\sqrt{2}}{\pi k} \sin \tfrac{\pi k}{2}.
\end{aligned}$$

iv. Students get full mark if they realize that, due to orthogonality of $\varphi(t)$, the corresponding wavelet is given by

$$\psi(t) = \sqrt{2}\sum_n (-1)^n g_0[1-n]\varphi(2t-n)$$

and if they write the corresponding expression

$$\psi(t) = \sum_n (-1)^n \frac{2}{\pi(1-n)} \sin\frac{\pi(1-n)}{2} \frac{\sin \pi(2t-n)}{\pi(2t-n)}.$$

Answers

4. (a) No. This is verified simply by noting that $\langle g_0[n], g_0[n-2] \rangle \neq 0$.

   (b) The filter-bank that we can obtain is biorthogonal. This means that $G_1(z) = z^{-1}H_0(-z)$. We therefore first focus on finding $H_0(z)$. We use the half band condition for that:

$$P(z) + P(-z) = 2$$

   with $P(z) = H_0(z)G_0(z)$. We pick $H_0(z)$ to be symmetric to guarantee that $G_1(z)$ is linear phase as requested in the question. We design $h_0[n]$ of length 5 and this leads to

$$P(z) = H_0(z)G_0(z) = \frac{1}{2\sqrt{2}}(z^{-1} + 2 + z)(az^{-2} + bz^{-1} + c + bz + az^2).$$

   Using the half-band condition we obtain

$$P(z) + P(-z) = \frac{1}{\sqrt{2}}(2az^{-2} + bz^{-2} + 2c + 2b + bz^2 + 2az^2) = 2$$

   which leads to the following two constraints $2a + b = 0$ and $c + b = \sqrt{2}$. Since we have three unknowns and only two equations, there is lack of uniqueness. We will arbitrarily choose a to give the same scaling as filter g_0, that is

$$a = \frac{1}{2\sqrt{2}}.$$

   Then

$$b = -\frac{1}{\sqrt{2}}$$

   and

$$c = \frac{3}{\sqrt{2}}.$$

   Finally, the synthesis highpass filter is:

$$G_1(z) = z^{-1}H_0(-z) = \frac{1}{2\sqrt{2}}z^{-1}(z^{-2} + 2z^{-1} + 6 + 2z + z^2).$$

   Please note that different choices of a would give different solutions, all valid. In particular

$$a = -\frac{1}{4\sqrt{2}}$$

   leads to the more commonly used synthesis filter

$$G_1(z) = z^{-1}H_0(-z) = \frac{1}{4\sqrt{2}}z^{-1}(-z^{-2} - 2z^{-1} + 6 - 2z - z^2).$$

   (c) i. $G_0(z) = \sqrt{2}\left(\frac{1+z^{-1}}{2}\right)\left(\frac{1+z}{2}\right)$, therefore $G_0(e^{j\pi}) = 0$ and $G_0(1) = \sqrt{2}$, so the necessary conditions for convergence are satisfied.

   ii. We can write $M_0(\omega)$ as follows

$$M_0(\omega) = \left(\frac{1 + e^{-j\omega}}{2}\right)^N R(\omega),$$

   where $R(\omega) = 1$ and $N = 2$. This means that $B = \max_\omega R(\omega) = 1$ and since $B < 2^{N-1}$ we know that the sufficient condition for $\varphi(t)$ to be continuous is satisfied.

Answers

iii. $G_0(z)$ has two zeros at $\omega = \pi$, therefore the analysis wavelet has two vanishing moments. Please note that this is true independently of the filter $G_1(z)$ we found in part (b).

Answers